



Petroleum Economics and Decision Analysis

Financial and Strategic Insights for Energy Decision-Makers



Petroleum Economics and Decision Analysis

The petroleum industry is complex, expensive, and uncertain. Success depends on smart choices and sound economic reasoning. As Warren Buffett said, “Risk comes from not knowing what you’re doing.” This course helps professionals make better decisions using clear insights—from economic models to risk and uncertainty analysis—built specifically for oil, gas, and energy investments.

By integrating financial principles, industry policies, and advanced scenario planning, participants gain the skills to make prudent project decisions. This course further prepares participants to tackle the challenges of decision making in the upstream oil and gas industry by providing them with skills in project decision analysis, economic evaluation, and policy analysis.

Software Requirements: The course uses Lumivero’s DecisionSuit Tools (formerly Palisade DecisionSuit Tools), including @risk and PrecisionTree. However, delegates do not need access to these platforms as other alternative solutions will be discussed during the course.



LOCATION

Dubai, UAE

Houston, USA

London, UK

Kuala Lumpur, Malaysia

LONDON PETRO ACADEMY
Empowering Professionals for a Sustainable Future



COURSE OBJECTIVES

This course highlights:

- Energy Commodity Market Economics: Understanding price, supply–demand dynamics, and investment trends.
- Financial Modeling & Valuation: Applying tools for assessing project profitability and economic feasibility.
- Risk & Uncertainty Management: Understanding, quantifying, and modeling uncertain factors.
- Decision Analysis Techniques: Use of Decision Trees, Monte Carlo Simulation and applications to Value of Information
- Energy Policy Impact: Evaluating impact of policies, ESG considerations, and geopolitical trends.
- Strategic Decision-Making: Optimizing investment portfolios and corporate planning strategies.

To succeed in the oil & gas industry is to create value. Learning about the subsurface challenges, along with the energy commodity market leads to insights that support good decision making. Good project decisions often lead to value creation.

At the end of this training course, you will learn to:

- Explain key aspects of the evolving energy landscape and its implications for oil and gas ventures.
- Summarise the fundamental principles of petroleum project economics and valuation techniques.
- Apply prudent business decisions to differentiate successful petroleum firms in an evolving global landscape.
- Analyse the impact of risk and uncertainty on management decision-making.
- Apply consistent financial valuation techniques to analyze petroleum projects.
- Use analytical tools and software for effective project evaluation and decision making.
- Evaluate the implications of uncertainty and risk on project feasibility
- Create decision support models that consider the influence of uncertain hydrocarbon prices and environmental considerations on value creation

Training Methodology

- Hands-on exercises using financial and statistical models.
- Real-world case studies covering successful and failed projects.
- Expert-led discussions on policy impacts and strategic decision-making.
- Collaborative scenario planning to develop strategic adaptability.
- Software-based simulation for uncertainty quantification.



Day 1

Industry Structure & Decision Context

- Global energy trends and geopolitical factors.
- Hydrocarbon industry: from subsurface to markets
- Industry competition and market forces: The emerging energy transition
- Value chain: economic principles underlying petroleum decision-making.
- Systematic approaches to managing uncertainty.
- Expected value concept in project decisions
- Introduction to decision analysis: framing choices and evaluating trade-offs in a real-world project.

Day 2

Financial Economics & Cash Flow Models

- Market-based approach to investment decision making
- Investment decision techniques: NPV, IRR, and Payback period.
- Sensitivity analysis in economics analysis of projects.
- Energy commodity markets: their insights for valuations.
- Commodity price forecasting and cash flow projections.
- **Case study:** Evaluating economic feasibility of Enceladus offshore oil field.

Day 3

Uncertainty Analysis & Quantitative Methods

- Uncertainty versus Risk
- Probabilistic reserves in valuations
- Risk classifications: market, political, technological, and operational.
- Sensitivity analysis in project assessments
- Decision trees and real options valuation in petroleum projects.
- **Case study:** Using Excel-based Monte Carlo models for Reservoir Uncertainty Modeling.

Day 4

Strategic Planning & Policy Economics

- Economic impact of energy transition policies.
- Petroleum fiscal regimes: taxes, royalties, and contracts.
- Policy incentives and tax structures shaping investment economics.
- Real-world corporate strategy responses to industry volatility.
- Scenario planning and stress testing for strategic adaptability.
- **Case Study:** Investing in Enceladus field development project.

Day 5

Practical Applications & Project Presentations

- Monte Carlo simulations for probability-based decision-making.
- Using spreadsheet add-ins: **Lumivero** @Risk and PrecisionTree
- Integrating economic, financial, and risk frameworks for decision-making.
- Advanced software models for petroleum economics.
- Evaluating long-term profitability: capital allocation and risk mitigation.
- Future trends: carbon markets, ESG considerations, and sustainable energy investments.
- **Capstone Project:** Presenting an investment recommendation based on course methodologies.

LUMIVERO

Take control of uncertainty with @RISK—the tool that empowers your team to quantify and mitigate risks with confidence, all within Microsoft Excel. Whether you're navigating financial markets, optimising operations, or ensuring regulatory compliance, @RISK transforms complex uncertainties into actionable insights through Monte Carlo simulation—enabling data-driven decisions that drive business success.



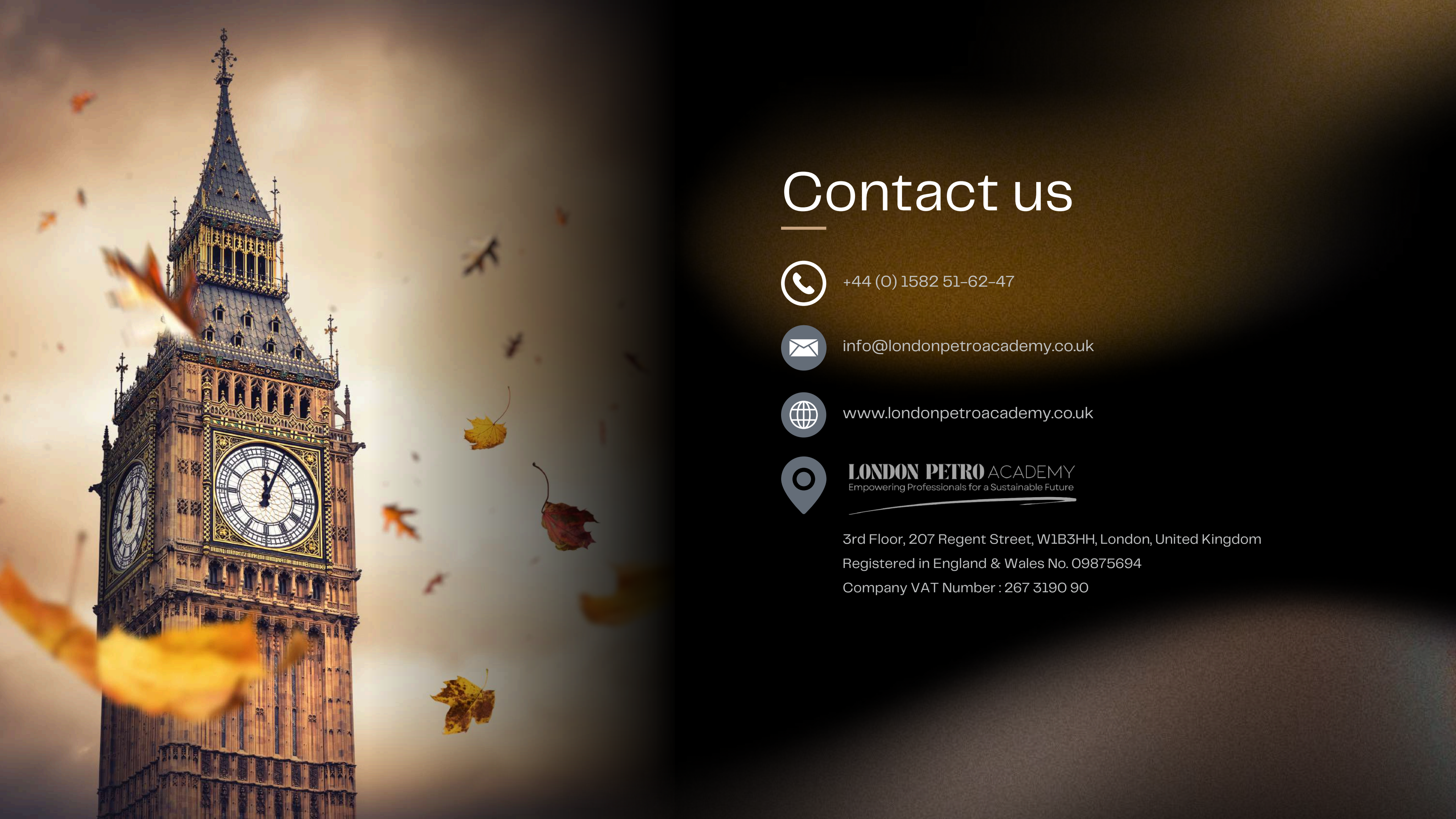
BABAK JAFARIZADEH

Dr. Babak Jafarizadeh is a senior consultant and engineering economist who views engineering as the pursuit of economic solutions to technical challenges in an uncertain world. With extensive experience in the oil and gas industry, he has served as a senior economic analyst supporting multi-million-dollar investment decisions for projects across the globe.

Since 2019, Dr. Jafarizadeh has worked as a senior consultant for leading training and consulting companies, while also holding academic appointments at several prestigious institutions — including Heriot-Watt University in the UK (Assistant Professor), the University of California, Berkeley (Lecturer at the Fung Institute for Engineering Leadership), and previously the University of Greenwich (Senior Lecturer). His earlier industry experience includes serving as a Senior Economic Analyst with Equinor (formerly Statoil) in Norway and the USA.

Dr. Jafarizadeh holds a PhD in Petroleum Engineering with a concentration in Investment and Decision Analysis from the University of Stavanger (in collaboration with the University of Texas at Austin), as well as an MSc in Financial Engineering and a BSc in Industrial Engineering. He is an active member of several professional associations, including the Society of Petroleum Engineers (SPE), the Institute for Operations Research and the Management Sciences (INFORMS), and the European Association of Geoscientists and Engineers (EAGE).

His teaching and research interests lie at the intersection of decision analysis and corporate finance, with particular applications in engineering and the energy industry. He has designed and delivered numerous professional training courses, such as Advanced Financial Modelling for Oil and Gas Professionals, Advanced Project Economics, Oil and Gas: Markets, Trading, Pricing and Economic Frameworks, and Energy Economics. Dr. Jafarizadeh is also the author of the book *Economic Decision Analysis (for Project Feasibility Studies)* and continues to contribute to both academia and industry through research, consulting, and executive education. His areas of expertise include petroleum economics, investment and decision analysis, project feasibility studies, corporate finance, and financial modelling for energy projects.



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